NSTS-37423

STS-88
SPACE SHUTTLE
MISSION REPORT

February 1999



**National Aeronautics and
Space Administration**

**Lyndon B. Johnson Space Center
Houston, Texas**

The RMS was repowered at 343:19:48 G.m.t. (05:11:13 MET) to support EVA 2. EVA 2 officially began when battery power was selected on the EMU's at 343:20:33 G.m.t. (05:11:58 MET). A nominal ODS depressurization in preparation for the EVA occurred at 343:20:13 G.m.t. (05:11:38 MET). All objectives of EVA 2 were successfully completed. An additional task completed on EVA 2 was the deployment of one of the Zarya module TORU antennas. The second TORU antenna was deployed during EVA 3. At approximately 344:00:52 G.m.t. (05:16:17 MET), a trunnion pin cover separated and floated away, thus only three of the four covers were installed on Unity. A new cover will be fabricated and installation of the fourth cover is planned for STS-96 (flight 2A.1). The EMUs performed nominally during the entire EVA. Only the RCS vernier thrusters were used for attitude control during the EVA. A nominal airlock repressurization following the EVA occurred at 344:03:35 G.m.t. (05:19:00 MET) to successfully conclude the 7-hour 2-minute EVA.

Following the EVA, the RMS was used to perform a survey of OSVS targets. Power was removed from the RMS at 344:05:49 G.m.t. (05:21:15 MET). The EMU water recharges were completed and a total of 7 pounds was added to both EMUs. The 20-hour EMU battery recharge was initiated at 344:04:59 G.m.t. (05:20:24 MET). The recharge on the pistol grip tool batteries and the EMU helmet light batteries was initiated at 344:05:16 G.m.t. (05:20:41 MET).

Node 1 of the ISS was ingressed at 344:19:54 G.m.t. (06:11:21 MET), and the FGB was ingressed at 344:21:11 G.m.t. (06:12:36 MET). The entire stack was repressurized to 14.7 psia between 344:22:36 G.m.t. (06:14:01 MET and 344:22:57 G.m.t. (06:14:21 MET).

The lights on panel A7 which indicate that payload bay camera B, an intensified television camera (ITVC), failed to illuminate. This condition is a known problem with this black-and-white camera as the data from the camera are interleaved in the video stream and the video switching unit (VSU) cannot always synchronize on these data. The history has shown that this condition occurs approximately 1/3 of the time, and a camera power-cycle usually clears the problem. However, in this case, the power was cycled at least eight times and the problem did not clear. A later power cycle did clear the problem, and the camera was left on for the remainder on on-orbit operations.

All transfer activities were completed, and all planned operations within the Functional Cargo Block (FGB) and Node 1 were also completed. A total of 1200 lb of hardware were transferred to the ISS and 335 lb were transferred from the ISS to the Orbiter. The FGB and Node 1 hatches were closed at approximately 346:00:25 G.m.t. (07:15:50 MET). The Orbiter was depressed to 10.25 psia in preparation for the third extravehicular activity at 346:01:06 G.m.t. (07:16:31 MET).

EVA 3 officially began when the EMU's were set to battery power at 346:20:33 G.m.t. (08:11:57 MET). A nominal Orbiter airlock depressurization in preparation for the EVA began at 346:20:17 G.m.t. (08:11:41 MET). The RMS was repowered at 346:19:51 G.m.t. (08:11:15 MET) to support the EVA . All objectives of EVA 3 were successfully completed, including the deployment of the second Zarya TORU antenna. The evaluation of the USA Simplified Aid for EVA Rescue (SAFER) which took place during this EVA achieved approximately 50 percent of its objectives due to uncertainties concerning the amount of gaseous nitrogen (GN_2) propellant that remained. The EMU's

EXTRAVEHICULAR OPERATIONS

The extravehicular activity (EVA) hardware performed satisfactorily throughout the mission and EVA operations were very successful with many tasks being completed that were to be completed on future mission EVA's.

The extravehicular mobility unit (EMU) checkout was performed at 339:06:19:36 G.m.t. (00:21:44:02 MET). All three suits performed nominally and were ready for the planned EVA's.

EVA 1 officially began when battery power was selected on the EMUs at 341:22:09 G.m.t. (03:13:34 MET). The foot restraint installation to the RMS striker bar was performed at 341:22:55 G.m.t. (03:14:20 MET). All objectives of EVA 1 were successfully completed, and the Zarya module and Unity module cable interfaces were verified with no anomalies noted. Another task completed during EVA 1 was the inspection of the undeployed TORU antennas on the Zarya module. From the inspection, it was determined that the antenna pyrotechnics have fired, but the antennas did not deploy.

At approximately 342:00:34 G.m.t. (03:16:59 MET), a slide-wire carrier became loose from the EV1 crewmember and floated away. At approximately 342:03:15 G.m.t. (03:18:40 MET), a work-site interface (WIF) socket and a retractable tether also became loose from the EV1 crewmember. Both of these items floated away from the Orbiter and ISS and caused no impact hazard.

The nominal airlock repressurization following EVA 1 began at 342:05:30 G.m.t. (03:20:55 MET), and the EMUs performed nominally during the entire EVA. A very successful 7 hour and 21 minute EVA was completed. Water recharging of the EMUs was completed at 342:07:07 G.m.t. (03:22:32 MET) in preparation for the next EVA, and the EMU battery recharging began at 342:07:40 G.m.t. (03:23:05 MET). EVA 1 was performed using only the RCS vernier thrusters for attitude control.

EVA 2 officially began when battery power was selected on the EMU's at 343:20:33 G.m.t. (05:11:58 MET). A nominal ODS depressurization in preparation for the EVA occurred at 343:20:13 G.m.t. (05:11:38 MET). All objectives of EVA 2 were successfully completed. An additional task completed on EVA 2 was the deployment of one of the Zarya module TORU antennas. The second TORU antenna was deployed during EVA 3. At approximately 344:00:52 G.m.t. (05:16:17 MET), a trunnion pin cover separated and floated away, thus only three of the four covers were installed on Node 1. A new cover will be fabricated and installation of the fourth cover is planned for STS-96 (flight 2A.1). The EMUs performed nominally during the entire EVA. A nominal airlock repressurization following the EVA occurred at 344:03:35 G.m.t. (05:19:00 MET) to successfully conclude the 7-hour 2-minute EVA.

EVA 3 officially began when the EMU's were set to battery power at 346:20:33 G.m.t. (08:11:57 MET). A nominal Orbiter airlock depressurization in preparation for the EVA began at 346:20:17 G.m.t. (08:11:41 MET). The RMS was repowered at 346:19:51 G.m.t. (08:11:15 MET) to support the EVA . All objectives of EVA 3 were successfully completed, including the deployment of the second Zarya TORU antenna.